2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA) | 978-1-6654-4337-1/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICMLA52953.2021.00189

Data Driven football scouting assistance with simulated player performance extrapolation

Shantanu Ghar
Computer Engineering
SIES Graduate School of Technology
Navi Mumbai, India
shantanughar11@gmail.com

Sayali Patil
Computer Engineering
SIES Graduate School of Technology
Navi Mumbai, India
sayalipatil631@gmail.com

Venkhatesh Arunachalam
Computer Engineering
SIES Graduate School of Technology
Mumbai, India
venkhatesharunachalam@gmail.com

***Abstract*—In club football, scouting is a crucial aspect of player recruitment, with elite football clubs investing millions of dollars in scouting and signing the best player for their team every year. Scouting requires great analytical and observational skills from the scout, to find the best player for any position in the team. A scout needs to analyze the player by watching his in-game actions, physical attributes and make a judgement on how the player might fit into the team. Every team has a formation, a style of play and a specific profile of player is required for a given position depending on the aforementioned factors. But scouts only watch a player play a few matches in person, and prepare their scouting report based on a player's performance in those matches. This process is flawed as the scout is expected to watch a few games and make estimates of the player's performance in a new team. The player statistics can help the scout in making better data-driven decisions.**

A player's career statistics can provide a picture of how the player performs individually, but they fail to predict player chemistry alongside a team. Misjudgement in scouting can lead to losses of millions of dollars to a club. We propose to solve this problem by utilising vast amounts of quantitative and qualitative player statistics (from 3+ sources), and by incorporating data science and machine learning algorithms to simulate real world performances of the team after the addition of the newly scouted player. We take into account specific player requirements and classify a player into one of our specific 15 player types, and use the team's formation and style of play to predict the players that will have the best chemistry with any given lineup, thereby facilitating scouts in making better decisions.

***Index Terms*—Data science, football, soccer, analytics, team chemistry, feature engineering, statistics, simulate, predict, ML, AI**

I. Introduction

Scouting is the process of analysing players from around the world and finding the most suitable player for a particular position in a team. This is the most crucial part in the football world where getting the right player is of utmost importance and has a huge contribution towards the success of the team. Every year a single football club spends millions of dollars on hiring players for their teams. Thus the player recruitment process needs close monitoring of the player in all his matches, and is analysed on the basis of his attributes, his strengths, weaknesses, tactical suitability and moreover his chemistry with the team. The people that carry out this task are called scouts. They are a group of people or a single person who is physically present to watch a few matches of a potential new player. Scouts observe the player, collect data, and evaluate the talent of footballers with an objective of signing them on a professional contract for their club .

Scouting is such an important aspect of the footballing world that a Professional Sports Football Scout Association was formed in 2013 to provide professional scouting courses [1]. This is a field where features of a player are analysed not only on the number of goals scored but also on the contribution of player actions towards the various facets of a team's overall play. Since the importance of scouting cannot be ignored, many attempts have been made by the professionals to automate this process, notably by Wyscout, ScoutMe, SciSports and the list goes on. These projects mainly include feature engineering, wherein the player attributes are compared for a particular position and the number of goals scored has a greater weight while finding the top players for a position. These systems are somewhat limited by the fact that they rely totally on the match statistics of players and fail to incorporate the qualitative aspects of the player, which is informally called the eye-test in football. These systems also overlook the style of play of a team and don't place emphasis on the coaches tactical requirement from a given player. Because, a player who is statistically great might be a good signing for one team, but might be a flop for a team with a different tactical setup. We work towards incorporating these crucial things in our system, to refine our ranking, thereby suggesting players that genuinely suit a given team.

Our system includes a huge amount of feature engineering, as each position in football requires a different set of player traits. These traits also depend on the formation and the style of play of the team. In this system, the user or the coach would feed the requirements of the team, such as the vacant position, specific player traits, the current team line-up, the playing style of the team and the team formation. This data is fed to our algorithm which in turn returns the list of top players according to the physical and tactical attributes required for that position. The list is then given as the input to our ML model which analyzes the team chemistry and returns the list of top N players. Thus, our system is proposed to give results similar to real life scenarios and helps the coach simulate a player in his team and predict how impactful the player could

978-1-6654-4337-1/21/$31.00 ©2021 IEEE
DOI 10.1109/ICMLA52953.2021.00189

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

be in the team.

II. The Problem

In the twentieth century, the rise of technology and the burgeoning internet infrastructure has resulted in football reaching millions of viewers in the click of a button. An explosion in the number of viewers has resulted in greater business opportunities, and consequently an exorbitant amount of money being plugged into football. The Premier league of England is the richest sports league in the world, and this explains the huge amount of money being spent on the recruitment of players.

All of this makes finding a player perfect for a position in the team critical, as there are heavy losses if the player does not fit the playing style of the team. The importance of scouting is not something that can be ignored, it needs a huge amount of research and analyses of players. To solve this problem, we propose an AI based solution to help in the "scouting" of players. Our proposed system will study the chemistry and playing style of the team, and find the best players to fit in a vacant position. It will then simulate the team's performance by placing the shortlisted player in that position, to help the coach zero in on the best candidate for his/her style of playing.

III. Proposed System

The proposed system is to create a web portal with an automated scouting system, i.e. our data science algorithm to suggest suitable players according to the requirements of the manager. The manager will provide his requirements as an input, and the output of our system will be a list of top N players which suit the given requirements.

A. Proposed system for existing drawbacks

Team chemistry is a very important factor in soccer. In soccer, the events are mostly based on offensive and defensive actions where communication plays a major role. If teams have good chemistry, they cooperate, understand and trust each other which leads to the success of the team. Currently, most of the existing systems do not take team chemistry into consideration for player performance prediction, thereby resulting in players being suggested only on the basis of their individual performances, in teams that are not tactically suitable. For example, if a world class player gets placed in the wrong team, it would take time for him to understand the team play, strategies etc, or to fit in the style of the team and to cooperate with the teammates. On the other hand even if an average player gets placed in a tactically suitable team, it would take no time for him to show his performance as he is already used to the game play of the current team, thereby helping both the player and the team to get the best out him, in a significantly less amount of time.

To tackle the aforementioned problems, we propose a 3 step framework which includes the following phases:

- Phase 1 - Preprocessing
- Phase 2 - Identify top players based on the player abilities and manager's tactical requirement (which includes parameters like vacant position, player attributes, playing style and formation).
- Phase 3 - Rank top N players from the above list based on team chemistry from 1 to N, with rank 1 being the best suited player and so on.

B. Components of the system

- Flask: Flask is a mirror web-framework written in python [6]. Our backend system i.e. machine learning model and preprocessing is done using python. Therefore, to connect our system with the network, and to make our codebase more uniform and easy to understand we chose flask as the web-framework in our system. Functionality of Flask
  1) Website built using Flask Jinja framework.
  2) RESTful API request handling.
  3) Integrating machine learning models and networks.
  4) Integration with Firebase Database.
- SPADL and VAEP Framework [3]: Our dataset is based on a collection of sequential actions performed by individual players. The data that we are using is present in the JSON format, which is difficult to visualise and process in the footballing context. Hence, we are using SPADL(Soccer Player Action Description Language), to convert the complex JSON format data to a readable tabular format and VAEP (Valuing Actions by Estimating Probabilities) to grade each player based on the in-game actions performed. The advantages of SPADL and VAEP are as follows :-
  1) Through SPADL, we can store every in-game action of an individual player, so that it is easy to understand and process the data for further analysis.
  2) Through VAEP, based on the action performed we can assign a positive score to each player when there's a team's chance of scoring and negative if there is not.

IV. Design and Methodology

A. Dataset

We have used data from various sources to cover all aspects of the player, which includes physical attributes, actions, tactics, etc. We have combined all the data into a single dataset which has all the data needed to analyse the player. The datasets that we have used in our system are as follows :-

- EA Sports FIFA Dataset
- FBref Dataset
- Wyscout Dataset

1) EA Sports FIFA Dataset [8]: The dataset contains 7 CSV files out of which 6 contain the player attributes and 1 contains the mapping of the different teams to their corresponding football league[7].

The 6 attribute based CSVs start with the FIFA 15 dataset and end with the FIFA 20 dataset, with the latter having 18,278 players and all of the CSVs having 104 columns representing the different attributes. The motivation behind choosing this

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

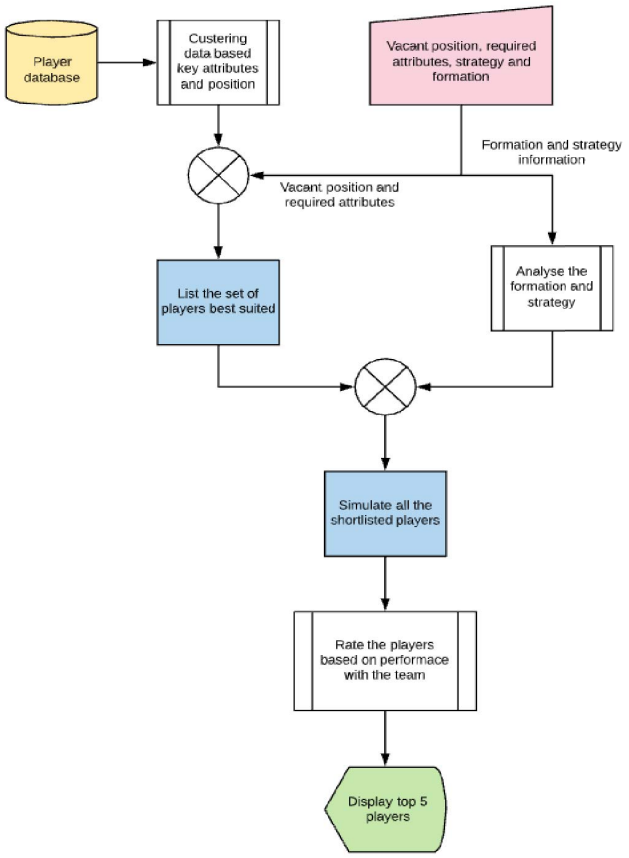


Fig. 1. System Flow Diagram.

dataset was that the historical data from 2015 to 2020 enables us to study the progression of players and the respective growth in their key attributes[1].

The dataset contains 104 different attributes for all players. These can be further categorized into the following types:

- Physical:- Height, Weight, Body Type, Preferred Foot, Weak Foot, Skills, Pace, Acceleration, etc.
- Attacking:- Shooting, Passing, Dribbling, Crossing, Finishing, Heading Accuracy, Volleys, etc.
- Skill Based:- Free Kick curve, Ball Control, Long Passing, Free Kick Accuracy, etc.
- Movement Based: - Movement, Agility, Movement Reactions, Sprint Speed, Balance,etc.
- Goalkeeping: Reactions, Diving, Reflexes, Handling, Positioning, etc.
- Defending: Man-marking, Standing tackle, Sliding tackle, Strength, Jumping, etc.
- Preferred Position: LS, ST, RS, LW, RW, CAM, CDM, CM, LCB, RWB,LB, RB etc.

All the above attributes are mapped to values ranging from 0 to 100, with 0 being the lowest score and 100 being the best possible score for the corresponding attribute. We chose this dataset as it is based on a game that is reviewed every year with updated stats based on the qualitative aspects of a player.

2) FBREF [9]: To get the real world analysis of every player, team and competition, we used the FBREF dataset. FBref provides statistics of every player based on their performance such as shooting, passes, fouls etc. For teams[8], FBref provides a dataset according to the competition such as English Premier League, National Team data, etc. The dataset contains 11 CSV files, some of them are listed below.

- Player Standard Stats - This CSV contains general information about the player i.e. age, team, nationality, etc.
- Player passing stats - This CSV contains player contributions to the team based on passing i.e. number of long passes, short passes, style of play, etc.
- Player Possession stats - This CSV contains player contribution to the team based on action i.e. touches, shots, etc.

3) WYSCOUT [10]: Wyscout provides a collection of soccer logs, containing all the events i.e. passes, shots, fouls, etc. The dataset contains various JSON files like events, competitions, matches, players, teams, referees and coaches[9].

- Matches - The match events JSON contains information like the time, outcome, players and other such characteristics of any match.
- Competition - This file contains a list of competition for a given area.
- Players - This JSON contains information about the player's full name, country, birth date and role.
- Teams - This JSON contains information about the team names, country, etc.
- Referees - This contains information about their name, nationality, etc.

B. Methodology

Our proposed methodology has 3 phases. Each phase mentioned below explains the rationale behind our data consolidation and processing.

1) Preprocessing: As mentioned earlier, we have used 3 different data sources in this project. To use all this data in the system, it is necessary to make the unstructured data uniform and structured in a way that data science algorithms can consume that data.

- Null Check - There were null values present in some rows and columns, and to ensure uniformity we dropped these rows, by eliminating records of players that had data less than a certain of number of appearances.
- Name Uniformity - Due to the usage of a combination of different datasets, some players had different names in different datasets. To map every player correctly, we added a player id and modified the raw data.
- Normalization - Some data was present in the form of percentiles and some in the form of integers with a variety of min and max ranges, so we normalized every numeric column to ensure values in a range of integers from 0 to 100. The reason behind choosing the range as 0 - 100 was the fact that the FIFA dataset was already in this range, and we wanted to evaluate other metrics in the

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

same way. This range has helped in making the overall rankings more lucid and convenient to understand, with each metric starting from the 0th percentile and going up to the 100th percentile.

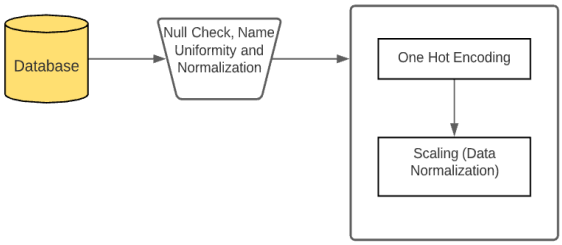


Fig. 2. Pre-processing Flow Diagram.

2) Attribute based filtering: In this module, we take into account the player type required by the manager for his team, based on the formation and playing style. In football, although the positions are broadly defined as goalkeeper, defender, fullback, midfielder, and forward, these positions are much more intricate than the above generalisations[1]. For e.g. a player deployed as a lone midfielder in front of the defenders can be tasked with distributing the ball with short passes and dictating the tempo of the game(e.g. Jorginho - Chelsea) or can be tasked with making recoveries and interceptions(e.g. N'Golo Kante - Chelsea). This shows that according to the formation, the playing style of the team, and the tactics of the manager, the same position can be occupied by players having completely contrasting attributes.

We took this into account and further classified the broad player positions of football into 15 specialised categories, namely :-

- FORWARDS :-
  1) Poacher :- A forward good in the opposition penalty area, and good at getting into attacking positions and finishing off chances without being industrious.
  2) Target Man :- This forward typically has a good height, is great at heading the ball and at hold-up play, and is the focal point of the passes of other teammates.
  3) Shadow Striker :- This is a player who is often in deeper positions compared to a traditional striker but uses late arrivals in the opposition's defensive third to shoot, and is usually paired with another striker.
  4) Mobile Forward :- This type of forward is given a license to roam around and press the opposition defenders to recover the ball and is quick with great stamina.
  5) Winger :- A winger is a wide forward who is great at crossing and drifting inside from the flanks. They are generally more creative than central strikers.
- MIDFIELDERS :-
  1) Box-To-Box :- A box to box midfielder is a midfielder capable of making runs from deep positions to the attacking third of the pitch. They have great all round defensive and offensive abilities.
  2) Deep Lying Playmaker :- This type of midfielder sits back close to the central defenders and is tasked with making frequent passes and running the game from deep positions.
  3) Attacking Midfielder :- This player is stationed between the forwards and the central midfielders, and is the creative fulcrum of the team.
  4) Central Defensive Midfielder :- This player lies in front of the central defenders and is great at tackling and intercepting with great stamina.
- FULLBACKS :-
  1) Wingback :- They are wide players stationed next to the midfielders, and are primarily supposed to help with the attacking, while going back to provide defensive cover.
  2) Fullback :- Traditionally, their task is to defend first, and assist in the attacks if the need arises.
- CENTRE BACKS:-
  1) Traditional Centre Half :- These are tall, physical defenders with great strength and tackling ability.
  2) Ball Playing Centre Back :- While having characteristics of a typical defender, these players also are great at ball distribution and start attacks from the deep.
- GOALKEEPERS:-
  1) Shot-Stopper :- These goalkeepers are great at stopping shots and have great reflexes.
  2) Sweeper :- These goalkeepers specialise in distribution, passing, and they aggressively roam in the box to clear out the ball if the need arises.

With all these player types defined, we take a combination of two datasets :

1) FBref
2) EA Sports FIFA dataset

To calculate the ranks of players for each of the aforementioned player types. The FBref data provides us with the real world stats like the number of goals, assists, tackles, interceptions, aerial duels won etc[8]. and the EA Sports FIFA dataset provides us with a qualitative measure of the players attributes like agility, heading, composure, aggression, strength etc[7].

We have consolidated the attributes required for each of the 15 playing styles, and we rank the players accordingly by combining data from the two datasets, and calculating a weighted, normalised score for each of these player types, thereby generating a list of the best players suiting that player type. In the next step, we pass this list to the chemistry analysis module, wherein these players are simulated in the team that needs a new player and the final list of best suited players is found[3].

C. Team Chemistry analysis

In this module, we focus on calculating team chemistry between 2 players. We are using a Wyscout dataset to process

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

each and every action performed by players during the entire season. An example of the raw Wyscout dataset in JSON format is given below.

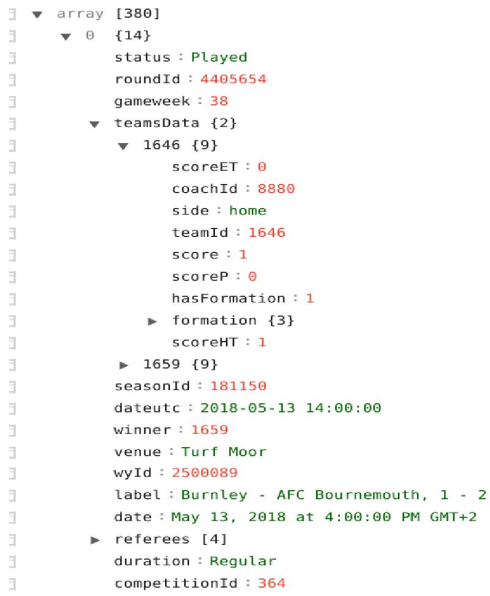


Fig. 3. Raw Wyscout data.

From Figure 3 we can judge that the JSON format is not human readable. Therefore, to make it human readable and understandable, we switched to SPADL - Soccer Player Action Description Language. SPADL is a language for describing player actions. Using SPADL we transformed the complex data of every in-game action into a simple format. [3]
Figure 4 is an example of the SPADL table format.

	game_id	period_id	seconds	team	player	start_x	start_y	end_x	end_y	actiontype	result
0	8657	2	2179	Belgium	Axel Witsel	37.1	44.8	53.8	48.2	pass	success
1	8657	2	2181	Belgium	Kevin De Bruyne	53.8	48.2	70.6	42.2	dribble	success
2	8657	2	2184	Belgium	Kevin De Bruyne	70.6	42.2	87.4	49.1	pass	success
3	8657	2	2185	Belgium	Eden Hazard	87.4	49.1	97.9	38.7	dribble	success
4	8657	2	2187	Belgium	Eden Hazard	97.9	38.7	105.0	37.4	shot	success

Fig. 4. SPADL Conversion

After structuring soccer data using SPADL, the next process is to grade the performance of each player based on the action performed. Actions in the dataset are categorized into 9 categories:- duel, foul, free kick, goalkeeper leaving the line, interruption, offside, pass, saves attempt and shot. Based on the actions performed by the player we assign grades to every player. The grade that we assign to every player is called the VAEP value (Valuing Actions by Estimating Probabilities). Based on the actions performed we modify VAEP values i.e. we add points to the VAEP value if the action contributes to scoring a goal, and decrease points if the action is based on decreasing the chance of scoring a goal.

VAEP values are increased and decreased based on the action. For example, a pass between 2 players will be penalized if the ball moves backward into a less favorable position. Whereas for an opposition player, points are awarded for moving the ball towards goal. In this way VAEP values are calculated for each player based on every action performed in their past matches [3].

	player_id	team_name	player	vaep_value	count
6	3009	France	Kylian Mbappé	3.788137	495
50	3308	England	Kieran Trippier	3.564976	690
122	4320	Brazil	Neymar	3.235207	749
17	3089	Belgium	Kevin De Bruyne	3.213972	716
92	3621	Belgium	Eden Hazard	3.092135	691
152	5186	Russia	Denis Cheryshev	3.022713	214
599	20004	France	Paul Pogba	3.011299	676
36	3244	England	John Stones	2.856670	937
352	5574	Germany	Toni Kroos	2.769772	647
121	4319	Uruguay	Edinson Cavani	2.769654	226

Fig. 5. Players with highest VAEP Values

In Figure 5 we have displayed top players with highest VAEP values based on the total actions performed in the overall season.

1) JI - Joint impact [4]: To calculate team chemistry we are using the JI metric, where we pair each player with another player and based on the actions of the pair, a positive or negative value is given. It is totally dependent on the VAEP value of each player action.

2) JOI - Joint Offensive Impact [4]: JOI is for a pair of players where if both the players are involved in an action which results in increasing a chance to score a goal then both of the players will receive credit for the pass. JOI for a pair of players in a match is calculated by summing up the VAEP values for the two constituting actions in an offensive sense.

$$JOI(p,q) = \sum VAEP(p,q) + \sum VAEP(q,p) \qquad (1)$$

JOI(p,q) is calculated by summing the VAEP values when p performs the first action and vice versa.

3) JDI - Joint Defensive Impact [4]: JDI is calculated for a pair of players where both the players are involved in an action which results in decreasing a chance to score a goal. To calculate JDI we need to calculate Offensive impact.
Offensive Impact (OI) - OI is the summation of VAEP values for the players action i.e. passes, take ons, shots etc.

$$JOI = \sum VAEP \qquad (2)$$

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

4) Expected Offensive Impact (EOI) [4]: EOI is calculated by taking the average of VAEP values in the matches of the same season that were played earlier.

5) Actual Offensive Impact (AOI) [4]: AOI is the summation of total VAEP value scored by the player in the current match. JDI is calculated by summing up the difference between expected offensive impact (EOI) and actual offensive impact (AOI).

$$JDI = \sum EOI - \sum AOI \tag{3}$$

6) Team Chemistry: Using JOI and JDI we can get insights of the team chemistry, but this will work only if a pair of players has played frequently. To overcome this situation we include a machine learning model to predict the VAEP value for new players based on their past performances. We have used a regression based algorithm which is explained in the next section, to predict Player Chemistry via a VAEP value by combining the JOI and JDI values of all pairs of players from the existing team with the new player. We declare the top 5 players with maximum VAEP value that are best suited for the team.

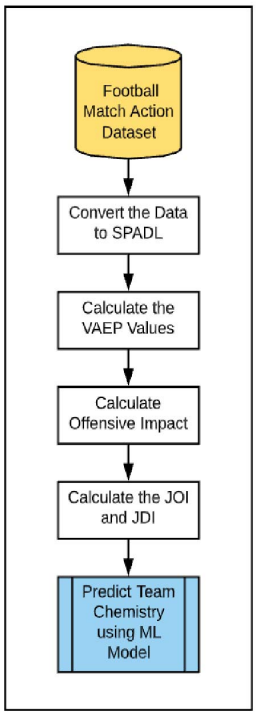


Fig. 6. High Level Overview of the System.

D. Algorithmic Implementation

The phase-wise algorithmic implementation is as follows :-

1) Preprocessing and data consolidation:-
   - One Hot Encoding :- Machine learning models and other data related functions cannot work with categorical data. As our dataset had a lot of categorical data (for e.g. formations, style of play, etc.), we had to convert the categorical columns into a format that was acceptable by the processing algorithms. The reason we chose the One Hot encoder instead of the LabelEncoder provided by sklearn was that the LabelEncoder is susceptible to establishing an uncalled hierarchy within the encoded columns, which is not the case with the One Hot Encoder.
   - Scaling :- We used the MinMaxScaler to standardise the player data from FBref.com to ensure that the algorithms had a uniform range to operate on. For instance, the number of recoveries ranged from 0-4000 in a season, but the aerials won ranged from 0-500 for a given team. The scaler scaled every column to a predefined range to make processing easier, while preserving the outliers.
2) Algorithm selection :-
   We used a machine learning algorithm to predict the chemistry of new players with the target team players. Initially, we took logistic regression as a baseline followed by random forest regression. Both logistic and random forest with 100 trees provided almost similar results. Later, on altering number of trees i.e gradually increasing to 500, random forest showed improvement in accuracy. Also, random forest works well with unbalanced and non linear data, and our final dataset contained independent features which made random forest more superior compared to linear regression based algorithms. Next, using random forest regression we predict JOI and JDI which enables us to get team chemistry insights.
3) Data preparation for training :-
   Our dataset contains performance values of 511 players. We divided the dataset into training and testing data where we used 80% of the dataset for training and 20% for testing.
   The input features of our system are as follows :-
   - `JOI_x and JOI_y`
   - `role_x and role_y`
   - `offensive_value_x and offensive_value_y`
   - `defensive_value_x and defensive_value_y`

   Here x and y determines the first and second player respectively.
4) Algorithm evaluation:- The metrics used for random forest regression evaluation are mean absolute error, mean squared error and root mean squared error.

   When using `n_estimators` (no. of trees) = 100, we got an RMS value of 0.42 and when using the number of trees as 500, we got the RMS value of 0.41 which was slightly better compared to the earlier result.

   Next, we evaluated the same scenario with 1000 trees, and there was no change in the RMS value, as seen in

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

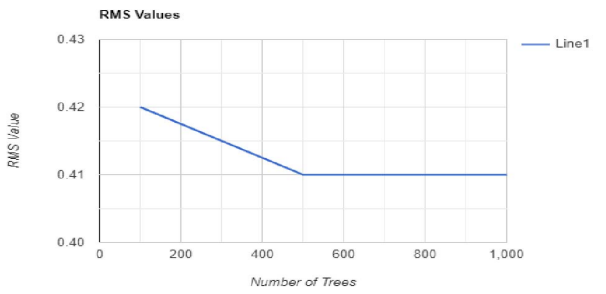


Fig. 7. RMS values over 100 - 1000 Trees.

Figure 7. Therefore, we fixed the number of trees to 500.

5) Results:-
We used our method to find the top 10 best suited players for a given team and then used the team chemistry simulation to shortlist the final top 5 players from the intermediate result. The example we considered for the input to our model was as follows: we tried to find the best forward, with the specific sub-type of "Poacher", for the team Arsenal F.C., which deploys a 4-3-3 formation and a predominantly high press style of play. Figure 8 shows the intermediate result consisting of top 10 players for the given inputs. Later these 10 players are simulated for team chemistry with the squad of Arsenal F.C. and the final top 5 players are given as output. Here, our algorithm predicted Harry Kane will be a good match, having the highest team chemistry score compared to other 4 players. And the fact that Harry Kane is a great poacher is corroborated by a 2016 news article [11]. Also, Wilfred Zaha is a player suggested by our model in Figure 8, who was actually linked with a move to Arsenal FC [12], thus showing the similarity of the predictions with real world events.

Figure 9 shows the results of our machine learning model, which predicts the top 5 players with the highest team chemistry [4].

E. Activity Flow

Here is a flowchart of the entire flow of selecting parameters and getting the top players suited for a particular team, style of play, and formation.

Step 1 Select the team for which a new player is to be brought in. This lineup is used later while simulating the performance of the intermediate list of players, to calculate the final list of players.

Step 2 Select the preferred formation for which the player must be suitable. The system keeps this in mind while generating the intermediate list, and prioritizes players that have played in that formation.

Step 3 Select the playing style of the team. e.g. High Press, Counter Attacking, Possession based, Low Block, etc.

	team	player_id	name	role_x	offensive	defensive	vaep_value
0	Tottenham	15808	Harry Kane	FW	16.03	-0.6000	15.97
1	Liverpool	832	Roberto Firmino	FW	8.90	-0.5000	8.40
2	Leicester	1002	Jamie Vardy	FW	8.00	-0.5000	7.50
3	Chelsea	12323	Romelu Lukaku	FW	5.90	-0.6400	5.26
4	Manchester City	3389	Leroy Sane	FW	6.00	-0.7676	5.24
5	Wolverhampton	4870	Raul Jimenez	FW	6.00	-0.7800	5.22
6	Manchester City	1024	Raheem Sterling	FW	5.80	-0.7700	5.03
7	Crystal Palace	2568	Wilfried Zaha	FW	5.50	-0.6000	4.90
8	Manchester City	9076	Sergio Aguero	FW	5.30	-0.5000	4.80
9	Chelsea	76	Alvaro Morata	FW	5.30	-0.5000	4.80

Fig. 8. Intermediate Results.

	team	player_id	name	role_x	offensive	defensive	vaep_value
0	Tottenham	15808	Harry Kane	FW	16.03	-0.6000	15.97
1	Liverpool	832	Roberto Firmino	FW	8.90	-0.5000	8.40
2	Leicester	1002	Jamie Vardy	FW	8.00	-0.5000	7.50
3	Chelsea	12323	Romelu Lukaku	FW	5.90	-0.6400	5.26
4	Manchester City	3389	Leroy Sane	FW	6.00	-0.7676	5.24

Fig. 9. Players with highest team chemistry.

Step 4 Select the broad vacant position for which a player is to be brought in. (e.g. Forward, Midfielder, Defender, Fullback, Goalkeeper)

Step 5 Select the specific player type within the broad playing position. (e.g. For a forward, select one of Poacher, Target man, Winger, Shadow Striker, etc.)

Step 6 Generate an intermediate player list for the given specifications. This ranked list contains the best players for the given position, and with the players that have a history of playing in the formations and tactical systems specified in the aforementioned steps.

Step 7 Take the intermediate list generated in Step 6, and calculate the JOI and JDI of each player in the list with

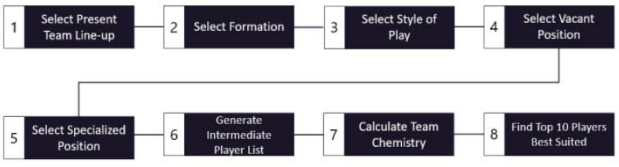


Fig. 10. Activity Flow.

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.

the current team lineup. The intermediate rankings are now recalculated and the final ranked list of the best suited players for the given inputs is shown.

CONCLUSION

Our system takes into account a multitude of statistics of every player's all round game, and combines the quantitative and qualitative aspects to generate a list of players that are of a very specific type and fit in a specific style of playing. We then use this list to simulate the impact that the player will have on the current team, thereby giving the manager an educated estimate of what he/she can expect from the newly recruited player, which can potentially save the club a huge amount of money compared to traditional scouting methods which operate without any extrapolation of the incoming player's current performance in a new team.

ACKNOWLEDGMENT

We are thankful for the support of our mentor Prof. Preeti Godabole, as her guidance was instrumental in helping us pull off this project. The authors thank FBref, Wyscout and Kaggle for the data used in this paper and the authors of the paper "Actions Speak louder than Goals", for their useful insights that proved to be helpful.

REFERENCES

[1] The Professional Football Scouts Association, "Elite Player Performance Plan". Available: https://www.pfsa.org.uk/

[2] Bradley, "Predicting Football Matches using EA Player Ratings and Tensorflow" , *https://towardsdatascience.com/*, Jul. 23, 2018. [Online] Available: https://towardsdatascience.com/predicting-premier-league-odds-from-ea-player-bfdb52597392

[3] T Decroos , L Bransen, JV Haaren ,J Davis (2019) Actions speak louder than goals: Valuing player actions in soccer. *In: Proceedings of the 25th ACM SIGKDD International Con-ference on Knowledge Discovery and Data Mining*, KDD 2019,August 4-8, 2019., pp 1851–1861.

[4] Lotte Bransen, Jan Van Haaren,"Player Chemistry: Striving for a Perfectly Balanced Soccer Team".

[5] Nazim Razali , Aida Mustapha , Faiz Ahmad Yatim , Ruhaya Ab Aziz ,"Predicting Player Position for Talent Identification in Association Football", *IOP Conference Series: Materials Science and Engineering*, Vol 226, 6–7 May 2017.

[6] Flask Tutorials,Available:https://flask.palletsprojects.com/en/1.1.x/tutorial/

[7] Andre Brener, "Selecting the best lineup for Argentina using AI and FIFA18", *medium.com* Jun 21, 2018[Online]. Available : https://medium.com/@andrebrener/selecting-the-best-lineup-for-argentina-using-ai-and-fifa18-4c776e5a655b

[8] Aman Shrivastava, "FIFA 18 Complete Player Dataset" , *kaggle.com*, Oct 30,2017[Online]. Available : https://www.kaggle.com/thec03u5/fifa-18-demo-player-dataset

[9] "2020-2021 Premier League Stats" , *kaggle.com*. Available : https://FBref.com/en/comps/9/Premier-League-Stats

[10] Wyscout, Available : https://wyscout.com/football-data-api/

[11] Articles by Agence France-Presse, "From goalkeeper to goal poacher: Harry Kane's rise to England's attack spearhead", *firstpost.com*, May 31, 2016[Online]. Available: https://www.firstpost.com/sports/from-goalkeeper-to-goal-poacher-harry-kanes-rise-to-englands-attack-spearhead-2808396.html

[12] Jordan Harris, "Report: Arsenal want Wilfried Zaha this summer; Crystal Palace exit inevitable", *https://tbrfootball.com/*. Available: https://tbrfootball.com/report-arsenal-want-wilfried-zaha-this-summer-crystal-palace-exit-inevitable/

Authorized licensed use limited to: VIT University- Chennai Campus. Downloaded on August 23,2026 at 17:28:58 UTC from IEEE Xplore. Restrictions apply.